

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

|                  |                                                                                                               |                   |                                                |
|------------------|---------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------|
| Course Code:     | CSA1715                                                                                                       | Course Title:     | Artificial Intelligence                        |
| CO Assessed:     | CO1 – Apply (BL3)                                                                                             | Assessment:       | Assessment Tool 1 – Analytical Problem Solving |
| Weightage:       | 35% of CIA                                                                                                    | Duration / Marks: | 60 mins                                        |
| CO1 Statement:   | Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3) |                   |                                                |
| Student Name:    | G Sai Sudhabdi                                                                                                | Reg. No.:         | 192472210                                      |
| Section / Batch: | CSE-AI / 2034                                                                                                 | Date:             | 03-08-2026                                     |

### Code of Conduct

I\_(G. Sai Sudhabdi / 192472219)\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Sai Sudhabdi

### Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evaluation page at last).
- Duration: 1 hour (15 minutes per question).

| Section / Topic                                | Focus                                                                                          | Question Nos. |
|------------------------------------------------|------------------------------------------------------------------------------------------------|---------------|
| 1. Foundations and History of AI               | Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI     | Q3            |
| 2. Intelligent Agents: Agents and Environments | Agent function and program, PEAS description, task environment properties, agent architectures | Q1, Q2        |
| 3. Concept of Rationality                      | Performance measure, percept sequence, expected utility, rational versus omniscient agent      | Q1, Q4        |

### Question 1: Two-Room Vacuum World

*Topic: Agents and environments, concept of rationality | 15 minutes*

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

**Performance measure over a lifetime of  $T = 6$  time steps:** +1 for every clean square at the end of each time step, and a penalty of  $p$  for every Left or Right move (take  $p = 1$  unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

- (a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**
- (b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for  $p = 1$ . **[5 Marks]**
- (c) Trace Agent Y in the same way and find its total score for  $p = 1$ . **[4 Marks]**
- (d) Derive the total score of X and of Y as functions of  $p$ . For which values of  $p$  is Y strictly better than X? **[3 Marks]**
- (e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

## Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

**Difficulty score convention:** for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

| Property              | Easier value (0) | Harder value (1)             |
|-----------------------|------------------|------------------------------|
| Observability         | Fully observable | Partially observable         |
| Number of agents      | Single agent     | Multi-agent                  |
| Determinism           | Deterministic    | Stochastic                   |
| Episodic / Sequential | Episodic         | Sequential                   |
| Static / Dynamic      | Static           | Dynamic (semi-dynamic = 0.5) |
| Discrete / Continuous | Discrete         | Continuous                   |

- (a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**
- (b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**
- (c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**
- (d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

### Question 3: Turing Test Data Across the History of AI

*Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes*

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

| System    | Era / technique                            | Interrogations | Correctly identified |
|-----------|--------------------------------------------|----------------|----------------------|
| ELIZA     | 1960s, keyword pattern matching            | 50             | 47                   |
| Chatbot B | 2010s, scripted rules plus statistical NLP | 50             | 36                   |
| System C  | 2020s, large neural language model         | 50             | 29                   |

**Reference:** Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

- (a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**
- (b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**
- (c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**
- (d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

## Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$U = 100 - T - 20 \cdot L$$

where  $T$  is the travel time in minutes and  $L = 1$  if the taxi arrives after the 55-minute deadline, otherwise  $L = 0$ .

| Route         | Normal time (min) | Probability of disruption | Time if disrupted (min) |
|---------------|-------------------|---------------------------|-------------------------|
| R1: Highway   | 30                | 0.30 (accident)           | 70                      |
| R2: City road | 45                | 0.10 (signal failure)     | 60                      |
| R3: Ring road | 50                | 0 (never disrupted)       | 50                      |

- (a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**
- (b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**
- (c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**
- (d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**
- (e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

## Rubrics for Calculation

| Criteria                         | Weightage  | Level 4 – Exemplary (100% of marks)                                               | Level 3 – Proficient (75%)                | Level 2 – Developing (50%)                 | Level 1 – Beginning (25%)                     |
|----------------------------------|------------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|
| <b>Problem Understanding</b>     | <b>20%</b> | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| <b>Method Selection</b>          | <b>20%</b> | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |
| <b>Solution Process</b>          | <b>25%</b> | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              |
| <b>Accuracy of Calculation</b>   | <b>20%</b> | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |
| <b>Interpretation of Results</b> | <b>15%</b> | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |

| Q. No. / Criteria | Problem Understanding (4) | Method Selection (4) | Solution Process (5) | Accuracy of Calculation (4) | Interpretation of Results (3) | Sub. Total (20) | Total % |
|-------------------|---------------------------|----------------------|----------------------|-----------------------------|-------------------------------|-----------------|---------|
| 1                 |                           |                      |                      |                             |                               |                 |         |
| 2                 |                           |                      |                      |                             |                               |                 |         |
| 3                 |                           |                      |                      |                             |                               |                 |         |
| 4                 |                           |                      |                      |                             |                               |                 |         |
| <b>Total</b>      |                           |                      |                      |                             |                               |                 |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

## 1. Two-Room Vacuum World

### (a) PEAS Description [4 Marks]

#### Performance Measure:

- Reward: +1 per clean square at the end of each time step over  $T = 6$  time steps.
- Penalty: -p penalty for every movement action ('Left' or 'Right').
- Total Score Formula:  $\sum_{t=1}^{\{6\}} \text{Clean Squares}_t - p * \text{Movement Penalty}_t$

#### Environment:

- Two discrete locations: Square A and Square B.
- Initial State: Agent is at A; both A and B are 'Dirty'.
- Deterministic dynamics: Once a square is cleaned, dirt never reappears.

#### Actuators:

- Operations available: 'Suck', 'Right', 'Left', 'NoOp'.

#### Sensors:

- Percept tuple: '[Current Location, Local Cleanliness State]'.
- Values: Location  $\sum \{A, B\}$ , State  $\sum \{\text{Clean, Dirty}\}$ .

### (b) Trace for Agent X (Simple Reflex Agent, $p = 1$ ) [5 Marks]

Agent X decides its action strictly based on the current percept. It cannot remember past states or visited locations.

- Initial State: Location = A, Squares Clean = 0 (A is Dirty, B is Dirty).

| Time Step (t) | Percept    | Action Chosen | Reason for Action                                 | Clean Squares at End of Step | Move Penalty (p=1) | Step Score |
|---------------|------------|---------------|---------------------------------------------------|------------------------------|--------------------|------------|
| 1             | [A, Dirty] | Suck          | Current square is Dirty<br>□ Suck                 | 1 (A clean)                  | 0                  | +1         |
| 2             | [A, Clean] | Right         | Current square is Clean<br>□ Move to other square | 1 (A clean)                  | 1                  | 0          |
| 3             | [B, Dirty] | Suck          | Current square is Dirty<br>□ Suck                 | 2 (A & B clean)              | 0                  | +2         |
| 4             | [B, Clean] | Left          | Current square is Clean<br>□ Move to other square | 2 (A & B clean)              | 1                  | +1         |
| 5             | [A, Clean] | Right         | Current square is Clean<br>□ Move to other square | 2 (A & B clean)              | 1                  | +1         |

| Time Step (t) | Percept    | Action Chosen | Reason for Action                              | Clean Squares at End of Step | Move Penalty (p=1) | Step Score |
|---------------|------------|---------------|------------------------------------------------|------------------------------|--------------------|------------|
| 6             | [B, Clean] | Left          | Current square is Clean □ Move to other square | 2 (A & B clean)              | 1                  | +1         |

Total Score for Agent X (p=1) = 1 + 0 + 2 + 1 + 1 + 1 = 5

**(c) Trace for Agent Y (Model-Based Reflex Agent with State, p = 1) [4 Marks]**

Agent Y maintains an internal state recording whether both squares have been cleaned.

- Initial State: Location = A, Internal State = 'BothClean = False'.

| Time Step (t) | Percept    | Action Chosen | Internal State Update / Reason           | Clean Squares at End of Step | Move Penalty (p=1) | Step Score |
|---------------|------------|---------------|------------------------------------------|------------------------------|--------------------|------------|
| 1             | [A, Dirty] | Suck          | A is Dirty □ Suck                        | 1 (A clean)                  | 0                  | +1         |
| 2             | [A, Clean] | Right         | A clean, BothClean=False □ Move to B     | 1 (A clean)                  | 1                  | 0          |
| 3             | [B, Dirty] | Suck          | B is Dirty □ Suck; sets BothClean = True | 2 (A & B clean)              | 0                  | +2         |
| 4             | [B, Clean] | NoOp          | B clean, BothClean=True □ NoOp           | 2 (A & B clean)              | 0                  | +2         |
| 5             | [B, Clean] | NoOp          | B clean, BothClean=True □ NoOp           | 2 (A & B clean)              | 0                  | +2         |
| 6             | [B, Clean] | NoOp          | B clean, BothClean=True □ NoOp           | 2 (A & B clean)              | 0                  | +2         |

Total Score for Agent Y (p=1) = 1 + 0 + 2 + 2 + 2 + 2 = 9

#### (d) Parametric Score Functions and Range of p [3 Marks]

Derivation of Scores as Functions of p:

##### 1. Agent X:

- Total clean-square rewards over 6 steps:  $1 + 1 + 2 + 2 + 2 + 2 = 10$
- Total moves executed: 4 ('Right' at  $t=2$ , 'Left' at  $t=4$ , 'Right' at  $t=5$ , 'Left' at  $t=6$ )
- Score Function:  $S_X(p) = 10 - 4p$

##### 2. Agent Y:

- Total clean-square rewards over 6 steps:  $1 + 1 + 2 + 2 + 2 + 2 = 10$
- Total moves executed: 1 ('Right' at  $t=2$ )
- Score Function:  $S_Y(p) = 10 - p$

Condition for Agent Y to be Strictly Better than Agent X:

Agent Y is strictly better than Agent X when  $S_Y(p) > S_X(p)$ :

$$10 - p > 10 - 4p$$

$$3p > 0$$

$$p > 0$$

For any positive move penalty ( $p > 0$ ), Agent Y strictly outperforms Agent X.

#### (e) Rationality Analysis and Interpretation [4 Marks]

##### 1. Rational Agent Identification

Agent Y is the rational agent.

##### 2. Why Percept History and Internal State Matter

- Definition of Rationality: An agent is rational if it chooses actions that maximize its expected performance measure given its percept history and built-in knowledge.
- The Limitation of Simple Reflex (Agent X): At steps 4, 5, and 6, Agent X receives the local percept '[Current Square, Clean]'. Because it has no memory, it cannot distinguish between:
  - ✓ "I just cleaned the first square and need to explore the second," and
  - ✓ "Both squares are already clean, so I should stay put."

As a result, Agent X falls into an infinite oscillation loop ('Left'  $\leftrightarrow$  'Right'), unnecessarily accumulating penalties of  $-4p$

Agent Y tracks cleaning history in memory, enabling it to recognize when both squares are clean. It then switches to NoOp, eliminating extra movement penalties and maximizing its overall performance score.

## 2. Task Environment Analysis and Agent Selection

### (a) Environment Classification and Justification [9 Marks]

#### E1: Autonomous Taxi Driving

- Observability: Partially observable (1) — The agent cannot see through buildings or detect traffic around distant corners.
- Number of agents: Multi-agent (1) — Operates alongside other drivers and pedestrians who are active decision-makers.
- Determinism: Stochastic (1) — Traffic behavior, weather, and pedestrian actions are non-deterministic.
- Episodic / Sequential: Sequential (1) — Current driving decisions (e.g., changing lanes) directly affect future traffic states.
- Static / Dynamic: Dynamic (1) — The environment continuously changes while the agent deliberates.
- Discrete / Continuous: Continuous (1) — Speed, steering angle, and spatial coordinates change continuously.

#### E2: Part-Picking Robot

- Observability: Fully observable (0) — Camera captures the entire workspace and part state on the belt.
- Number of agents: Single agent (0) — The robot operates independently without other active decision-making agents.
- Determinism: Deterministic (0) — Given part location and grip action, the state transition is fully predictable (assuming no hardware failure).
- Episodic / Sequential: Episodic (0) — Picking each part is an independent episode that does not affect future parts.
- Static / Dynamic: Semi-dynamic (0.5) — Belt moves while thinking, but the environment structure does not change independently.
- Discrete / Continuous: Continuous (1) — Joint angles, velocity, and visual positioning are continuous variables.

#### E3: Chess-Playing Agent (with Clock)

- Observability: Fully observable (0) — The complete board state is visible at all times.
- Number of agents: Multi-agent (1) — Competes directly against an opponent player.
- Determinism: Deterministic (0) — Every valid move leads to a precise, known successor board state.
- Episodic / Sequential: Sequential (1) — Opening moves strongly influence endgame position and strategy.
- Static / Dynamic: Semi-dynamic (0.5) — The game state remains still during a turn, but elapsed clock time penalizes long deliberation.
- Discrete / Continuous: Discrete (0) — Moves, piece positions, and board squares are strictly discrete.

### (b) Difficulty Score Computation and Ranking [4 Marks]

Difficulty Score =  $\sum$  Property Scores

- Score E1 (Taxi):  $1 + 1 + 1 + 1 + 1 + 1 = 6.0$
- Score E2 (Robot):  $0 + 0 + 0 + 0 + 0.5 + 1 = 1.5$
- Score E3 (Chess):  $0 + 1 + 0 + 1 + 0.5 + 0 = 2.5$

Ranking (Hardest to Easiest): E1 (6.0) > E3 (2.5) > E2 (1.5)

### (c) Agent Architecture Selection [4 Marks]

Hardest Environment (E1: Autonomous Taxi)

- Architecture: Utility-based Agent or Utility-based Learning Agent.
- Justification: E1 is stochastic, sequential, dynamic, and multi-agent. The agent must trade off conflicting goals (safety, speed, comfort, passenger cost) under uncertainty. A utility function maps states to real numbers, enabling optimal decision-making when trade-offs and probabilities are involved.

#### Easiest Environment (E2: Part-Picking Robot)

- Architecture: Model-based Reflex Agent or Simple Reflex Agent.
- Justification: E2 is fully observable, deterministic, and episodic. A model-based reflex agent uses camera inputs to track moving parts across conveyor belt frames, matching simple condition-action rules ('if defect detected at position (x,y) -> actuate picker') without needing complex lookahead search.

#### **(d) Interpretation of Environment Properties vs. Agent Complexity [3 Marks]**

- Environment properties dictate how much information an agent must maintain and compute.
- Partially observable and dynamic environments force the agent to maintain an internal state (world model) and deliberate quickly under time constraints.
- Stochastic and sequential properties require probabilistic lookahead and planning across future states rather than simple lookup rules.
- Continuous environments demand function approximation and numerical optimization instead of basic discrete search tables.

### 3. Turing Test Data Across the History of AI

#### (a) Correct-Identification Rate and Fooling Rate [6 Marks]

$$\text{Correct - identification rate} = \frac{\text{Correctly identified as machine}}{50} \times 100$$

The **fooling rate** is the percentage of times the machine was mistaken for a human:

$$\text{Fooling rate} = 100 - \text{Correct - identification rate}$$

| System    | Correctly identified | Correct-identification rate | Fooling rate |
|-----------|----------------------|-----------------------------|--------------|
| ELIZA     | 47/50                | 94%                         | 6%           |
| Chatbot B | 36/50                | 72%                         | 28%          |
| System C  | 29/50                | 58%                         | 42%          |

**Interpretation:** System C has the highest fooling rate, meaning judges were most likely to mistake it for a human.

#### (b) Turing's 70% criterion and distance from guessing

Turing's prediction says that after five minutes, an average interrogator would have **no more than a 70% chance of correctly identifying the machine**.

Therefore, I compare each system's correct-identification rate with **70%**.

- **ELIZA:**  $94\% > 70\% \rightarrow$  **Does not satisfy Turing's criterion.**
- **Chatbot B:**  $72\% > 70\% \rightarrow$  **Does not satisfy Turing's criterion**, although it is very close.
- **System C:**  $58\% < 70\% \rightarrow$  **Satisfies Turing's criterion.**

The random-guessing level is **50%**. The distance from this level is:

$$\text{Distance from guessing} = \text{Correct - identification rate} - 50\%$$

| System    | Correct-identification | Distance from 50% guessing | Turing criterion |
|-----------|------------------------|----------------------------|------------------|
| ELIZA     | 94%                    | +44 percentage points      | No               |
| Chatbot B | 72%                    | +22 percentage points      | No               |
| System C  | 58%                    | +8 percentage points       | Yes              |

Thus, System C comes closest to the situation where judges are essentially unable to distinguish it reliably from a human. However, **58% is still above pure chance**, so it should not be interpreted as perfect human indistinguishability.

#### (c) Improvement in fooling rate

Fooling rates:

- ELIZA = **6%**
- Chatbot B = **28%**
- System C = **42%**

**ELIZA  $\rightarrow$  System C**

**Improvement in percentage points:**

$$42\% - 6\% = 36 \text{ percentage points}$$

**Improvement as a ratio:**

$$\frac{42}{6} = 7$$

So, System C's fooling rate is **7 times** that of ELIZA.

**Chatbot B  $\rightarrow$  System C**

$$42\% - 28\% = 14 \text{ percentage points}$$

Therefore, System C improves the fooling rate by **14 percentage points** over Chatbot B.

**(d) Interpretation: Acting humanly vs. thinking rationally**

The Turing Test primarily evaluates whether a machine can **act humanly** during a conversation. A high fooling rate means that the system produces responses that judges perceive as human-like. It does **not automatically demonstrate genuine thinking, understanding, consciousness, or rational decision-making**.

For example, the **Chinese Room argument** by John Searle challenges the idea that producing appropriate language responses proves understanding. A system could manipulate symbols according to rules and still lack actual semantic understanding of what those symbols mean.

Therefore, System C's **42% fooling rate** shows a strong improvement in human-like conversational behavior, but it does not prove that System C is actually thinking or acting rationally. It only demonstrates performance on one behavioral test.

**Rubric-based conclusion**

- **Problem Understanding:** Correctly distinguishes the Turing Test's focus on human-like behavior.
- **Method Selection:** Uses the "acting humanly" approach and Chinese Room argument.
- **Solution Process:** Connects fooling rate to conversational behavior rather than assuming intelligence.
- **Accuracy:** Correctly interprets the 42% fooling rate.
- **Interpretation:** Clearly explains why imitation of human conversation is not proof of genuine thought or rationality.

#### 4. Rational Route Choice under Uncertainty

The utility function is:

$$U = 100 - T - 20L$$

where:

- $T$  = travel time in minutes
- $L = 0$  if the taxi arrives by the deadline
- $L = 1$  if the taxi arrives after 55 minutes

The important point is that **arriving late produces an additional penalty of 20 utility units.**

##### (a) Utility for each route and outcome

###### R1: Highway

Normal travel time = 30 minutes.

Since  $30 \leq 55$ :

$$L = 0 \\ U = 100 - 30 - 20(0) = 70$$

**R1 normal: U = 70, on time.**

If disrupted, travel time = 70 minutes.

Since  $70 > 55$ :

$$L = 1 \\ U = 100 - 70 - 20(1) \\ U = 100 - 70 - 20 = 10$$

**R1 disrupted: U = 10, LATE.**

###### R2: City road

Normal travel time = 45 minutes.

Since  $45 \leq 55$ :

$$U = 100 - 45 - 20(0) = 55$$

**R2 normal: U = 55, on time.**

If disrupted, travel time = 60 minutes.

Since  $60 > 55$ :

$$U = 100 - 60 - 20(1) = 20$$

**R2 disrupted: U = 20, LATE.**

###### R3: Ring road

Travel time = 50 minutes and there is no disruption.

Since  $50 \leq 55$ :

$$U = 100 - 50 - 20(0) = 50$$

**R3: U = 50, on time.**

##### Summary

| Route | Outcome | Time | Late? | Utility |
|-------|---------|------|-------|---------|
|-------|---------|------|-------|---------|

|    |        |        |    |    |
|----|--------|--------|----|----|
| R1 | Normal | 30 min | No | 70 |
|----|--------|--------|----|----|

|    |           |        |     |    |
|----|-----------|--------|-----|----|
| R1 | Disrupted | 70 min | Yes | 10 |
|----|-----------|--------|-----|----|

|    |        |        |    |    |
|----|--------|--------|----|----|
| R2 | Normal | 45 min | No | 55 |
|----|--------|--------|----|----|

|    |           |        |     |    |
|----|-----------|--------|-----|----|
| R2 | Disrupted | 60 min | Yes | 20 |
|----|-----------|--------|-----|----|

| Route | Outcome   | Time   | Late? | Utility |
|-------|-----------|--------|-------|---------|
| R3    | Normal    | 50 min | No    | 50      |
| R3    | Disrupted | —      | —     | —       |

R3 has no disruption, so its utility is always 50.

### (b) Expected utility and rational choice

For uncertain outcomes:

$$EU = \sum P(outcome) \times U(outcome)$$

#### R1

Probability of disruption = 0.30.

Therefore:

$$\begin{aligned} P(normal) &= 1 - 0.30 = 0.70 \\ EU(R1) &= (0.70)(70) + (0.30)(10) \\ &= 49 + 3 \\ EU(R1) &= 52 \end{aligned}$$

#### R2

Probability of disruption = 0.10.

$$\begin{aligned} P(normal) &= 0.90 \\ EU(R2) &= (0.90)(55) + (0.10)(20) \\ &= 49.5 + 2 \\ EU(R2) &= 51.5 \end{aligned}$$

#### R3

There is no disruption:

$$\begin{aligned} EU(R3) &= 1(50) \\ EU(R3) &= 50 \end{aligned}$$

### Comparison

| Route         | Expected Utility |
|---------------|------------------|
| R1: Highway   | 52               |
| R2: City road | 51.5             |
| R3: Ring road | 50               |

Therefore, the rational taxi agent chooses:

*R1: Highway*

because it has the **highest expected utility of 52**.

### (c) New radio alert: R1 disruption probability becomes 0.80

The new information changes the probability, not the utilities of the outcomes.

For R1:

$$\begin{aligned} P(disruption) &= 0.80 \\ P(normal) &= 0.20 \end{aligned}$$

Therefore:

$$\begin{aligned} EU(R1) &= (0.20)(70) + (0.80)(10) \\ &= 14 + 8 \\ EU(R1) &= 22 \end{aligned}$$

The other routes remain:

$$EU(R2) = 51.5$$

$$EU(R3) = 50$$

Therefore, after receiving the radio alert:

*Choose R2: City road*

The decision changes because the new percept makes the highway substantially riskier.

**(d) If the disruption actually occurred, was the agent irrational?**

In part (b), the rational agent selected **R1**, because:

$$EU(R1) = 52$$

Suppose the accident actually occurs.

The actual travel time is 70 minutes, so the taxi arrives after the 55-minute deadline.

Thus:

$$L = 1$$

and:

$$U = 100 - 70 - 20(1)$$

$$U = 10$$

The agent receives a utility of only **10**.

However, this **does not mean the agent was irrational**.

At the time of making the decision, the agent did not know whether the accident would occur. Based on the available information, R1 had the highest expected utility.

A **rational agent** chooses the action that maximizes expected performance using the information available at the time.

An **omniscient agent**, on the other hand, would know the actual future outcome. If it knew the accident was going to happen, it would avoid R1.

Therefore:

*The agent was rational, even though the outcome was unfavorable.*

Bad luck is not necessarily irrational decision-making.

**(e) Interpretation: percept sequence and performance measure**

Rationality depends on the **percept sequence** because an agent makes decisions based on what it has observed so far.

Initially, the agent knows that R1 has only a 30% disruption probability:

$$EU(R1) = 52$$

so R1 is the best choice.

After the radio alert, the agent receives new information indicating an 80% probability of disruption:

$$EU(R1) = 22$$

Now R2 becomes preferable.

The roads themselves have not changed, but the agent's **information about the environment has changed**. Therefore, a rational agent should update its decision.

The performance measure also matters. Here, the taxi is rewarded for **short travel time** but penalized for **missing the 55-minute deadline**. If the performance measure only considered travel time and ignored lateness, the preferred route could be different.